



ScholarNest

Spark Azure Databricks

Databricks Spark Certification and beyond

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Azure
Databricks



Absolute Beginner to Specialization in Apache Spark and Azure Databricks



Micro Project – Problem Statement

We have seen a Spark example earlier which gives us the following feel:

1. What do we do in the Spark application?
2. How does a typical Spark application code looks?
3. How to write and execute your Spark application?

Apache Spark is a data processing engine.
In a typical Spark application, we read data from a source, process it, and write or present the results.
And this course is all about learning how to do these three things - Read, Process and Write.
We have seen a sample code of the Spark application, where we learned how to write and execute Spark code.

It's time to start learning Spark Programming and develop your Spark coding skills. We will take a project-driven approach to learning things.

We have a micro project problem statement shown below.

I am giving you the Fire Calls-For-Service dataset.

Problem Statement

Data Set: Fire Department Calls for Service

URL: <https://data.sfgov.org/Public-Safety/Fire-Department-Calls-for-Service/nuek-vuh3>

What are the requirements

1. Load the given data file and create a Spark data frame.
2. Use the Spark data frame to answer the following questions.
 1. How many distinct types of calls were made to the fire department?
 2. What are distinct types of calls made to the fire department?
 3. Find out all responses or delayed times greater than 5 mins?
 4. What were the most common call types?
 5. What zip codes accounted for the most common calls?
 6. What San Francisco neighborhoods are in the zip codes 94102 and 94103
 7. What was the sum of all calls, average, min, and max of the call response times?
 8. How many distinct years of data are in the CSV file?
 9. What week of the year in 2018 had the most fire calls?
 10. What neighborhoods in San Francisco had the worst response time in 2018?

Reference: <https://data.sfgov.org/Public-Safety/Fire-Department-Calls-for-Service/nuek-vuh3>

This data set represents a response to calls at fire units. If we go to the above link, we can see the 34 column names available in this data set. You can also see the table preview in this page if you scroll down.

Fire Department Calls for Service

data.sfgov.org/Public-Safety/Fire-Department-Calls-for-Service/nuek-vuh3

Columns in this Dataset

Column Name	Description	Type
Call Number		Plain Text T
Unit ID		Plain Text T
Incident Number		Plain Text T
Call Type		Plain Text T
Call Date		Date & Time
Watch Date		Date & Time
Received DtTm		Date & Time

Show All (34)

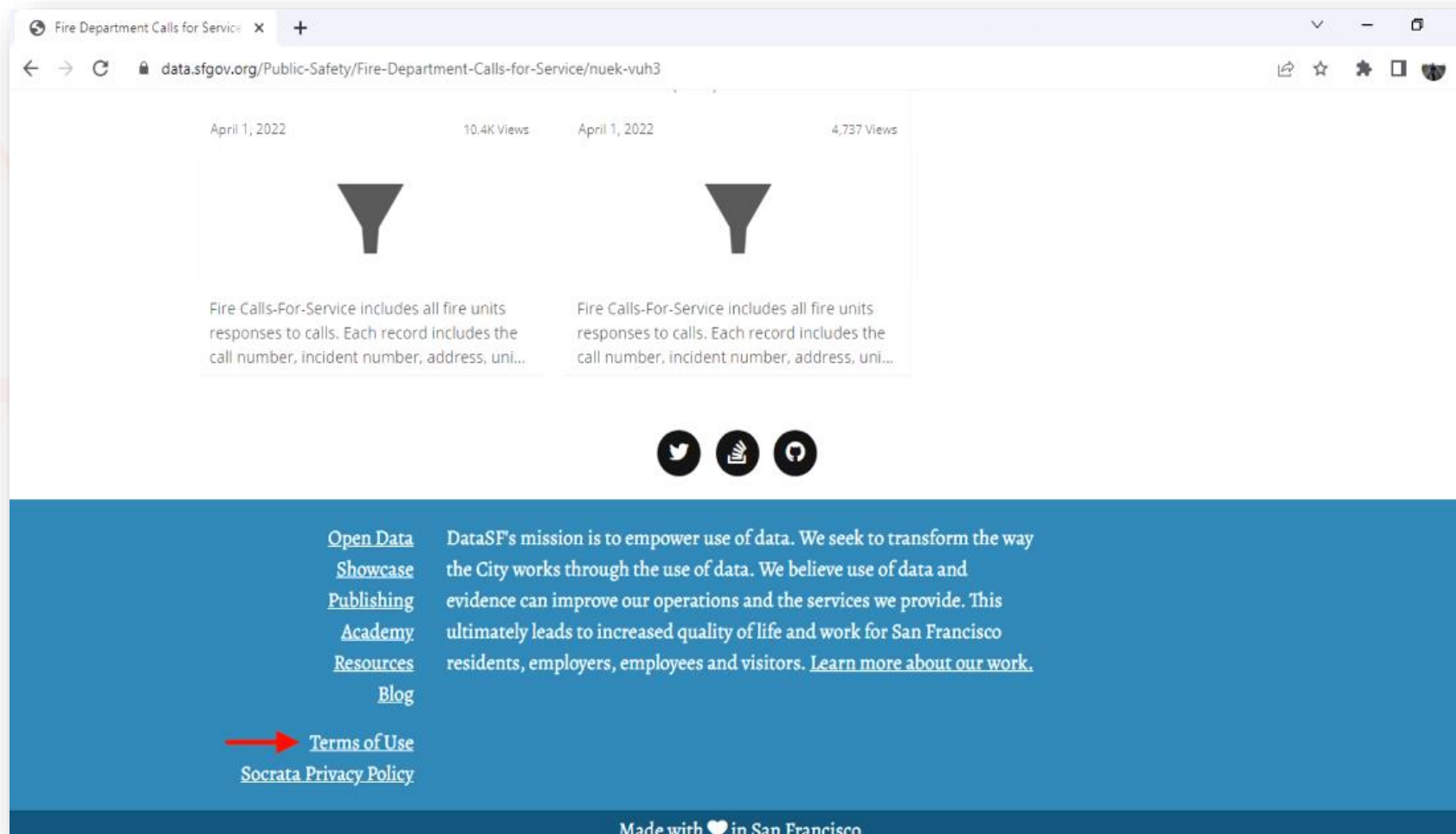
Table Preview

View DataCreate Visualization

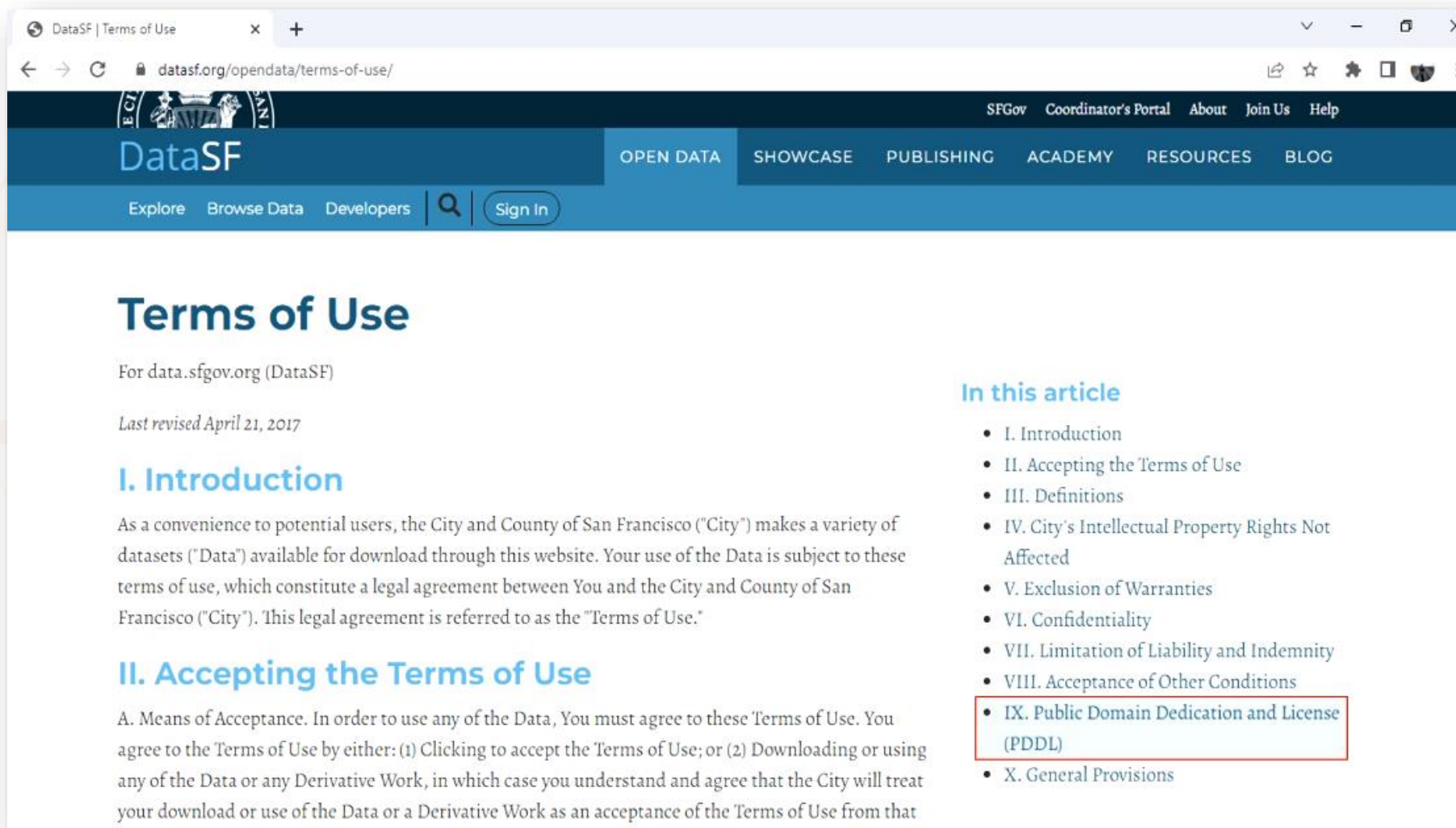
Call ...	Unit ID	Incid...	Call T...	Call ...	Watc...	Recel...	Entry...	Disp...	Resp...	On S...	Tran...	Hos
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Reference: <https://data.sfgov.org/Public-Safety/Fire-Department-Calls-for-Service/nuek-vuh3>

Scroll to the bottom of this page, you will see a link for terms of use, click that.



The data set is available under the PDDL terms. Click the PDDL link you see in the side panel.

A screenshot of a web browser displaying the 'Terms of Use' page for DataSF (data.sfgov.org). The browser's address bar shows the URL 'datasf.org/opendata/terms-of-use/'. The page features a dark blue header with the DataSF logo and navigation links like 'OPEN DATA', 'SHOWCASE', 'PUBLISHING', 'ACADEMY', 'RESOURCES', and 'BLOG'. Below the header, the main content area is white. The title 'Terms of Use' is prominently displayed in a large, bold, dark blue font. Underneath the title, it says 'For data.sfgov.org (DataSF)' and 'Last revised April 21, 2017'. The page is divided into sections: 'I. Introduction' and 'II. Accepting the Terms of Use'. The 'II. Accepting the Terms of Use' section contains a paragraph about the legal agreement and a sub-section 'A. Means of Acceptance'. To the right of the main text, there is a sidebar titled 'In this article' which lists ten items. Item 'IX. Public Domain Dedication and License (PDDL)' is highlighted with a red rectangular border. The browser's interface, including the back, forward, and refresh buttons, is visible at the top.

Terms of Use

For data.sfgov.org (DataSF)

Last revised April 21, 2017

I. Introduction

As a convenience to potential users, the City and County of San Francisco ("City") makes a variety of datasets ("Data") available for download through this website. Your use of the Data is subject to these terms of use, which constitute a legal agreement between You and the City and County of San Francisco ("City"). This legal agreement is referred to as the "Terms of Use."

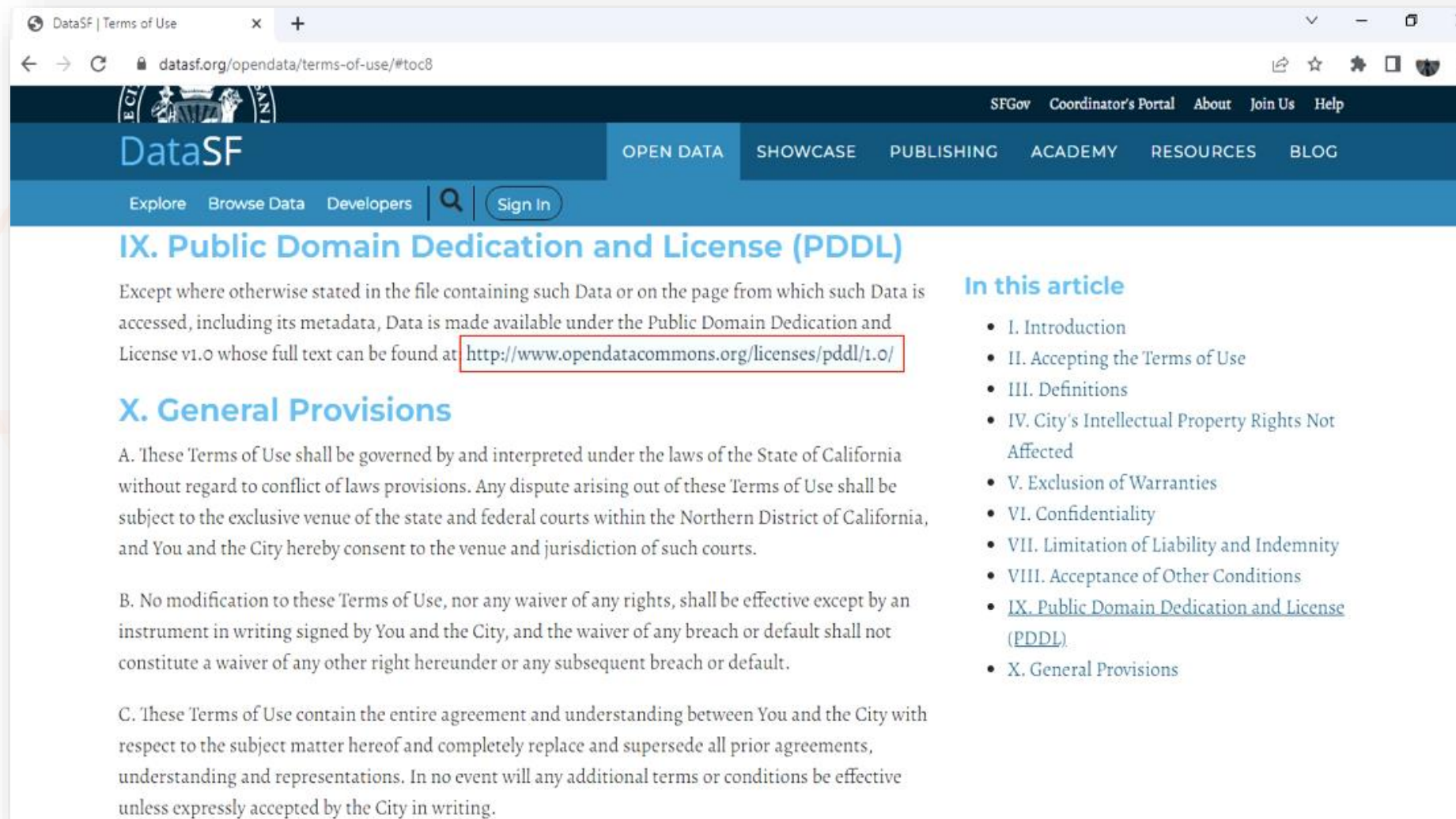
II. Accepting the Terms of Use

A. Means of Acceptance. In order to use any of the Data, You must agree to these Terms of Use. You agree to the Terms of Use by either: (1) Clicking to accept the Terms of Use; or (2) Downloading or using any of the Data or any Derivative Work, in which case you understand and agree that the City will treat your download or use of the Data or a Derivative Work as an acceptance of the Terms of Use from that

In this article

- I. Introduction
- II. Accepting the Terms of Use
- III. Definitions
- IV. City's Intellectual Property Rights Not Affected
- V. Exclusion of Warranties
- VI. Confidentiality
- VII. Limitation of Liability and Indemnity
- VIII. Acceptance of Other Conditions
- IX. Public Domain Dedication and License (PDDL)
- X. General Provisions

You will see the open data commons URL. Copy the URL and paste it into a new browser tab.



The screenshot shows a web browser window with the address bar displaying `datasf.org/opendata/terms-of-use/#toc8`. The page header includes the DataSF logo and navigation links: SFGov, Coordinator's Portal, About, Join Us, Help, OPEN DATA, SHOWCASE, PUBLISHING, ACADEMY, RESOURCES, and BLOG. Below the header, there are links for Explore, Browse Data, Developers, and a Sign In button. The main content area is titled "IX. Public Domain Dedication and License (PDDL)" and contains the following text:

Except where otherwise stated in the file containing such Data or on the page from which such Data is accessed, including its metadata, Data is made available under the Public Domain Dedication and License v1.0 whose full text can be found at <http://www.opendatacommons.org/licenses/pddl/1.0/>

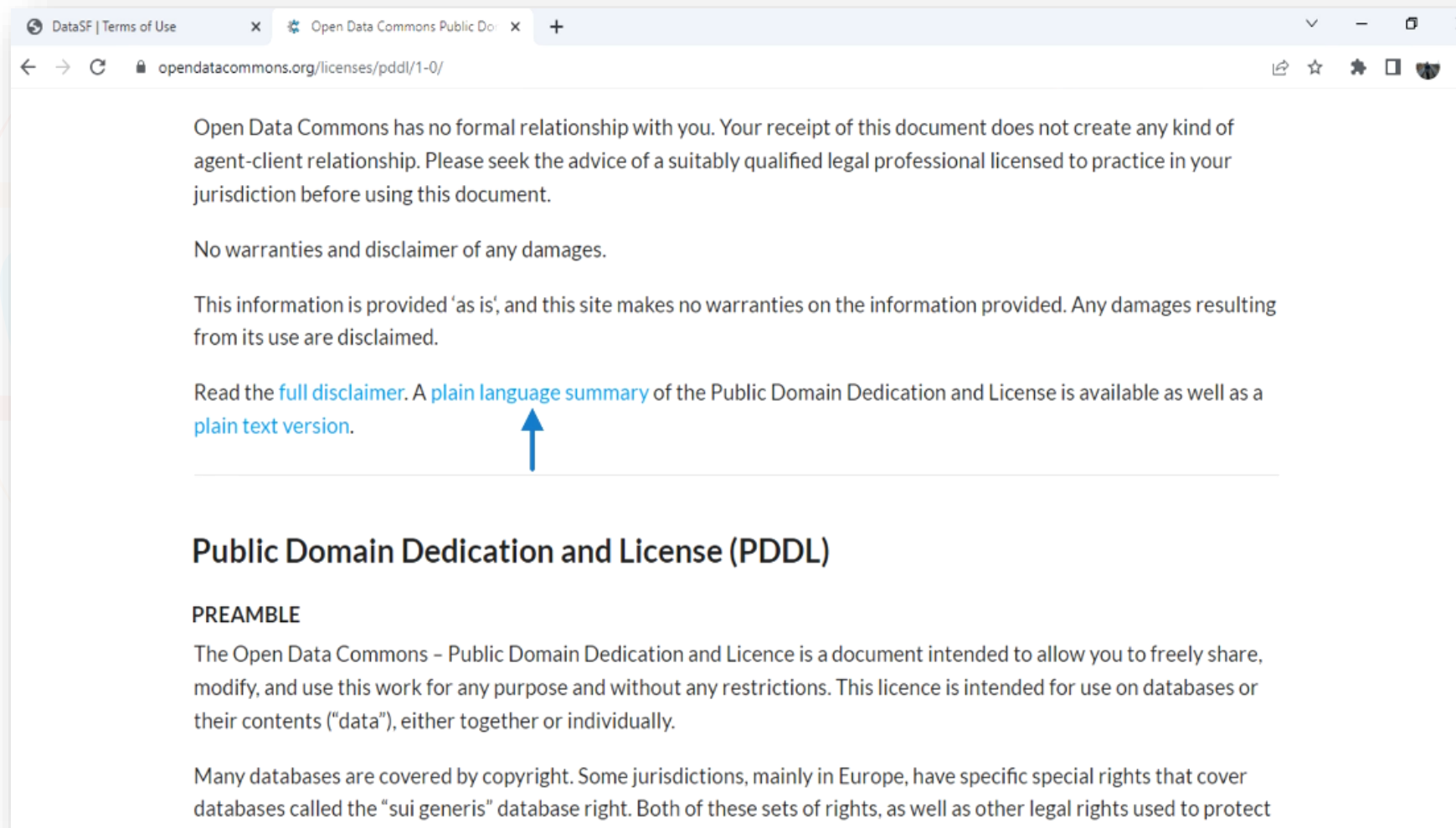
Below this text is the section "X. General Provisions" with three sub-sections:

- A. These Terms of Use shall be governed by and interpreted under the laws of the State of California without regard to conflict of laws provisions. Any dispute arising out of these Terms of Use shall be subject to the exclusive venue of the state and federal courts within the Northern District of California, and You and the City hereby consent to the venue and jurisdiction of such courts.
- B. No modification to these Terms of Use, nor any waiver of any rights, shall be effective except by an instrument in writing signed by You and the City, and the waiver of any breach or default shall not constitute a waiver of any other right hereunder or any subsequent breach or default.
- C. These Terms of Use contain the entire agreement and understanding between You and the City with respect to the subject matter hereof and completely replace and supersede all prior agreements, understanding and representations. In no event will any additional terms or conditions be effective unless expressly accepted by the City in writing.

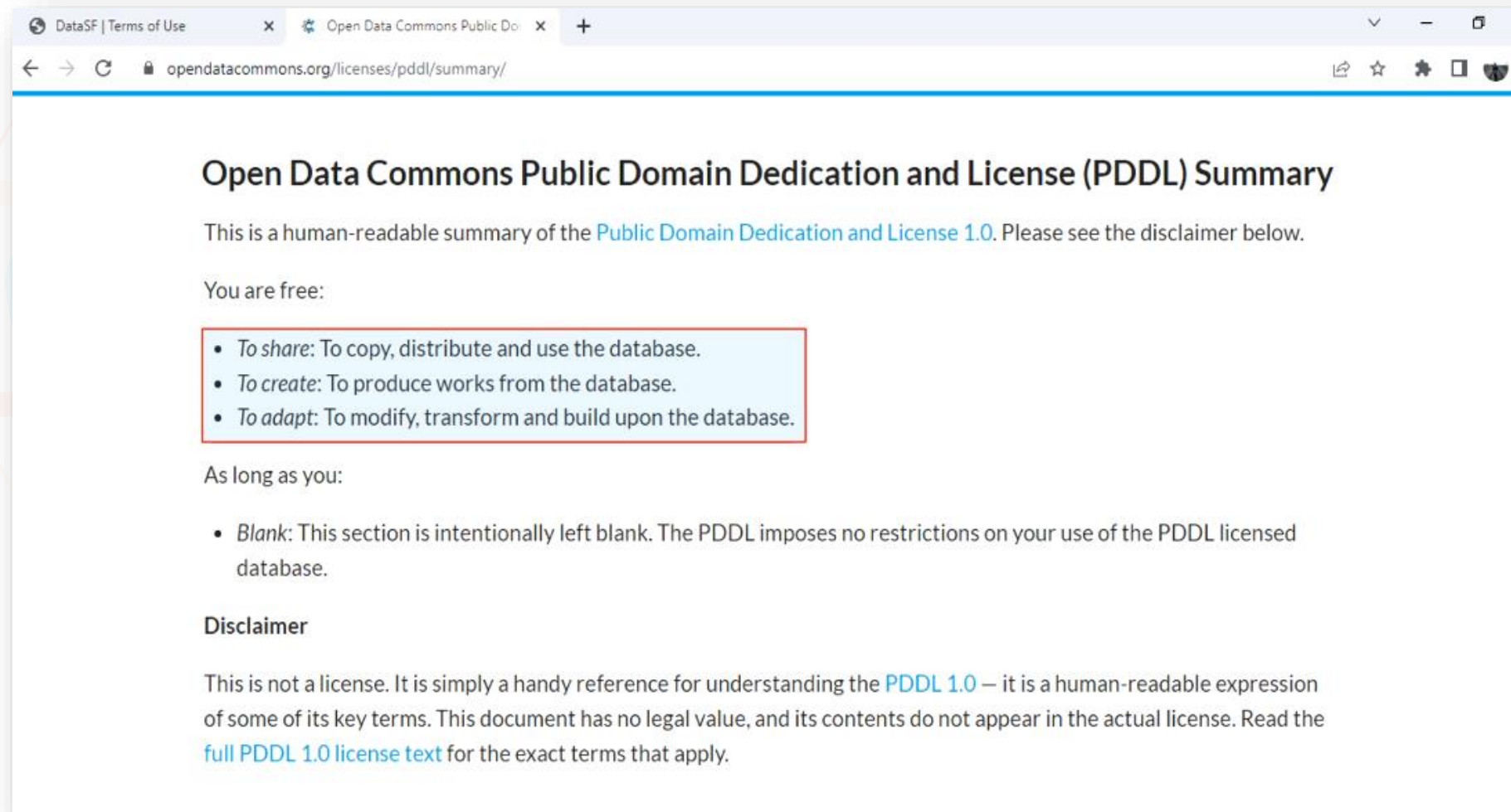
On the right side of the page, there is a section titled "In this article" with a list of links:

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If you go to the open data commons URL, and scroll down, you will see a plain-language summary. Click that.



So you are free to share, create and adapt this data set. It is essential to check the terms of use for any data set that you are using for your learning and experiments. I showed you how to check the license terms for the Fire Calls-For-Service dataset.



You are given a San Francisco fire call response data set. And you are asked to fulfil the following set of requirements

Problem Statement

Data Set: Fire Department Calls for Service

URL: <https://data.sfgov.org/Public-Safety/Fire-Department-Calls-for-Service/nuek-vuh3>

What are the requirements

1. Load the given data file and create a Spark data frame.
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 10. What neighborhoods in San Francisco had the worst response time in 2018?

We want to do only two things:

1. Load the data into a Spark Data frame
2. Query it to find the answers to the ten questions shown earlier.

The main objective of the Micro Project is to learn two things:

1. How to load data into Spark
2. How to query data in Spark.

We have two approaches to do it:

1. Using Spark SQL
2. Using Spark Dataframe API

While we work on the solution of the given 10 questions, we will learn the following concepts:

1. Using Spark SQL and DDL statements
2. Spark Data Frame and Data Frame Reader API
3. Working with CSV Files
4. Data Frame Transformations and Actions
5. How to read Spark API Documentation



Thank You
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